

CLASS 10 · ICSE · SELINA CONCISE MATHEMATICS · PYQ

# Chapter 8: Remainder and Factor Theorems

Previous year board questions, topic-wise: 14 questions

The tag after each question gives the board examination year and the marks. Solutions are in a separate PDF with the same question numbers.

## 1 The factor theorem

1. Which factor is common to  $x^2 - 4$  and  $x^3 - x^2 - 4x + 4$ ? (a)  $(x + 1)$  (b)  $(x - 1)$  (c)  $(x - 2)$  (d)  $(x - 4)$  **ICSE 2025, 1 mark**

## 2 Factorising a cubic polynomial

2.  $(x - 2)$  is a factor of  $2x^3 - 7x^2 + kx - 2$ . Find  $k$ , and then factorise the polynomial completely. **ICSE 2025, 3 marks**
3.  $(x - 1)$  is a factor of  $3x^3 + 8x^2 - 15x + k$ . Find  $k$ , and then factorise the polynomial completely. **ICSE 2024, 3 marks**
4. Use the factor theorem to factorise  $2x^3 - x^2 - 13x - 6$  completely. **ICSE 2023, 4 marks**
5. Use the factor theorem to factorise  $6x^3 + 17x^2 + 4x - 12$  completely. **ICSE 2020, 3 marks**
6. Use the factor theorem to show that  $(x - 2)$  is a factor of  $x^3 + x^2 - 4x - 4$ . Then factorise the polynomial completely. **ICSE 2019, 3 marks**
7. Use the remainder theorem to factorise  $2x^3 + 3x^2 - 9x - 10$ . **ICSE 2018, 3 marks**

## 3 The remainder theorem

8. What must be subtracted from  $x^3 + x^2 - 2x + 1$  so that the result is exactly divisible by  $(x - 3)$ ? (a)  $-31$  (b)  $-30$  (c)  $30$  (d)  $31$  **ICSE 2024, 1 mark**
9. What must be added to  $2x^3 - 3x^2 - 8x$  so that it leaves a remainder of 10 when divided by  $2x + 1$ ? **ICSE 2020, 3 marks**
10. Find the remainders when  $x^3 + (kx + 8)x + k$  is divided by  $x + 1$  and by  $x - 2$ . The two remainders add up to 1. Find  $k$ . **ICSE 2019, 3 marks**
11. What must be subtracted from  $16x^3 - 8x^2 + 4x + 7$  so that the result has  $2x + 1$  as a factor? **ICSE 2017, 3 marks**

## 4 Finding unknown constants

12.  $(x - 2)$  is a factor of  $x^3 - kx - 12$ . Find  $k$ . (a) 3 (b) 2 (c)  $-2$  (d)  $-3$  **ICSE 2023, 1 mark**
13. Find  $a$  if  $(x - a)$  is a factor of  $3x^3 + x^2 - ax - 81$ . **ICSE 2023, 4 marks**

14.  $(x + 2)$  and  $(x + 3)$  are factors of  $x^3 + ax + b$ . Find  $a$  and  $b$ . **ICSE 2018, 3 marks**

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